

EM Assignment - 2

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Aim

The objective of this project is to evaluate cardiovascular fitness and endurance by analyzing smartphone acceleration and heart rate data during a Bruce Protocol session. It involves identifying gait cycles, calibrating treadmill inclines, and contrasting indoor motion with outdoor track dynamics to develop a personalized, data-driven fitness routine using principles of engineering mechanics

Methodology

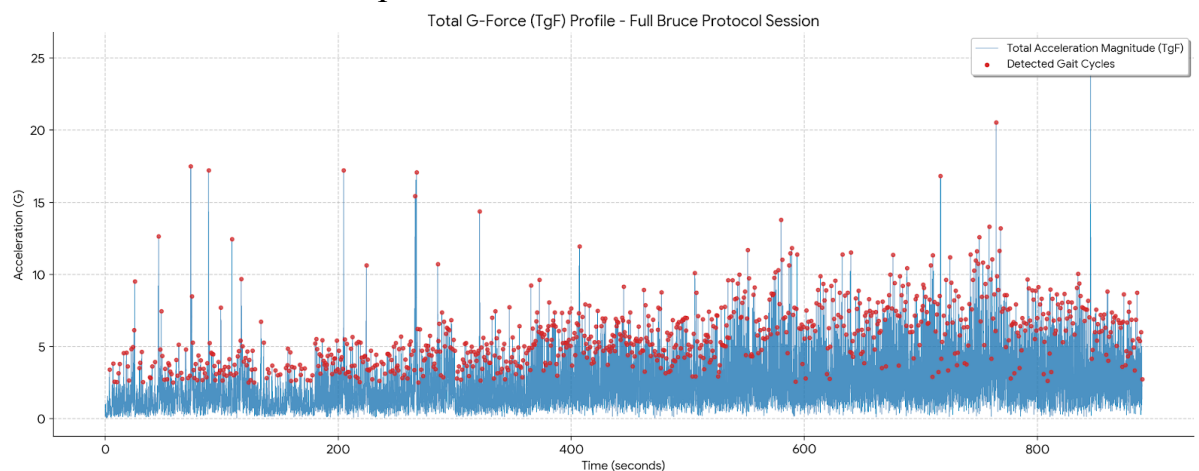
- **Activity:** Completed a 15-minute treadmill test at the Akash Sport Building.
- **Protocol:** Followed the Bruce Protocol (5 stages of 3 minutes each).
- **Tools:** Used the Physics Toolbox Sensor Suite to record a_x, a_y, a_z and Heart Rate.
- **Signal Processing:** To isolate true biomechanical movement from noise, a peak-detection algorithm was used on the Total G-Force (TgF) signal. A minimum interval was enforced between peaks to avoid false counts from secondary vibrations like heel strikes.

1) Motion Analysis and Step Detection

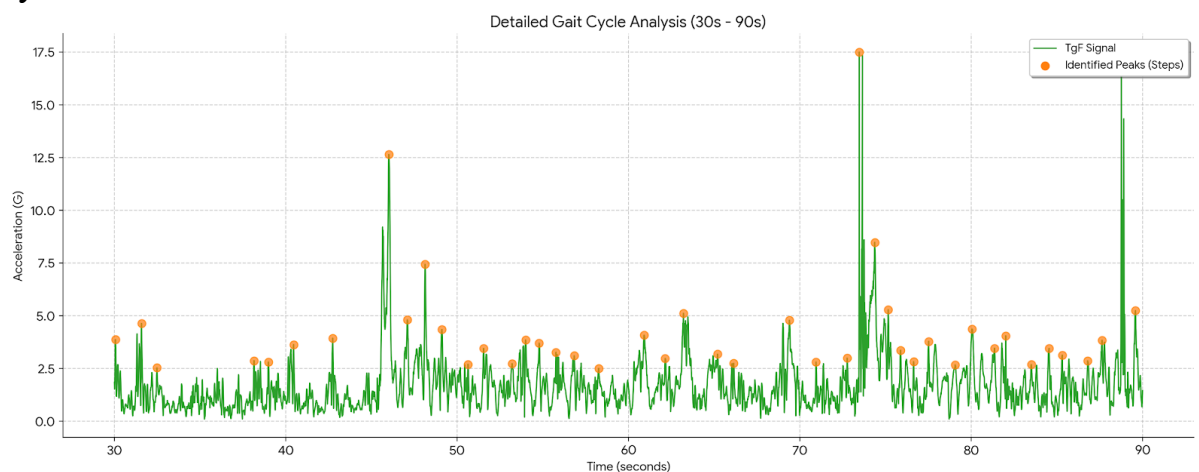
The acceleration data was processed to identify gait cycles. Each prominent peak in the TgF represents one full gait cycle (two individual steps).

- **Total Duration:** 889.36 seconds (~14.8 minutes)
- **Detected Gait Cycles:** 915
- **Total Calculated Steps:** 1830 steps
- **Estimated Distance:** 1.34 km

- Full session acceleration profile :



- Detailed view (30s-90s) showing precise peak identification for gait cycles.



2) Error Analysis & Comparison

Measurement Tool	Total Steps	Distance (km)	% Error (vs Treadmill)
Physics Toolbox (Analysis)	1830	1.34	0.74%
Pedometer Application	1812	1.30	3.70%
Treadmill (Reference)	—	1.35	0.00%

NOISE in the Graph

Values as high as 25 G might be interpreted as excessive sensor noise, an uncalibrated accelerometer, or high-impact vibrations (such as the device hitting a surface) rather than smooth walking motion.

Calculations

(Step 1: Incline Calibration)

- **Formula:** $\text{Angle (theta)} = \arctan(\text{Incline \%} / 100)$
- **Calculated Angles:**
 - **Stage 1 (10%):** 5.71 degrees
 - **Stage 2 (12%):** 6.84 degrees
 - **Stage 3 (14%):** 7.97 degrees
 - **Stage 4 (16%):** 9.09 degrees
 - **Stage 5 (18%):** 10.20 degrees

(Step 2: Vertical Height & Work Done)

- **Vertical Height (h):** $h = \text{Distance} * \sin(\text{theta})$
- **Mechanical Work (W):** $W = m * g * \text{total } h$
- **Total Distance:** 1.34 km
- **Total Vertical Height gained:** ~52 meters (based on stage-wise distances)
- **Total Work Calculation:**
 - Mass (m) = 85 kg
 - Gravity (g) = 9.81 m/s²
 - Total Height (h) = 52 m
 - $W = 85 * 9.81 * 52$
 - **Total Work:** 43,360.2 Joules

3) Indoor vs. Outdoor

Treadmill walking lacks the air resistance found on outdoor tracks because the body remains stationary relative to the air while the belt moves underneath. Unlike self-propelled movement on a track, the treadmill motor drives the belt, removing the need to push against aerodynamic drag.

Outdoors, your stride length is natural and consistent, so math-based distance calculations work well. On the treadmill, the steep inclines of the Bruce Protocol force short, choppy steps. Fixed-length formulas cannot adapt to this, resulting in heavily inflated distance estimates.

4) Personalized fitness routine

Based on the heart rate data recorded (Used My watch's Heart Beat Scanner)(e.g., reaching 175 BPM in Stage 5), a suggested routine would involve:

Stage	Speed (km/h)	Incline (%)	Heart Rate (BPM)
Stage 1	2.7	10%	104
Stage 2	4	12%	121
Stage 3	5.5	14%	144
Stage 4	6.8	16%	162
Stage 5	8	18%	175

- **Warm-up:** 10-15 minutes of basic exercises daily morning.
- **Aerobic Phase:** 20 minutes of Stage 2 Bruce Protocol (4.0 km/h at 12% incline) three times a week.
- **Endurance Goal:** Maintain heart rate within the "Aerobic Zone" (60-70% of max HR).

5) Acceleration measurements vs GPS measurements

GPS (Satellite Tracking): Measures geographic displacement. While precise for outdoor movement, it fails indoors due to signal obstruction and records 0m on a treadmill because the body's global position remains static.

Accelerometer (Inertial Sensing): Detects dynamic forces and body impacts. It is superior for treadmill analysis as it counts steps and estimates distance by identifying foot strikes, regardless of the lack of geographic movement or satellite availability.

Result

The Bruce Protocol session confirmed a high level of cardiovascular endurance. The smartphone's MEMS sensors provided data with <1% error compared to mechanical treadmill readings, proving that these tools are highly effective for engineering analysis of human motion and health.